# Ernest Skrzypczyk

*Bachelor of Science in Electrical Engineering - Control Engineering and Robotics*

*15 Rue Fresche Blance*
*44300 Nantes*
*France*
*+48 515 89 08 19*
*ernest.skrzypczyk@gmail.com*

## Education

2015–current **Double Masters Degree**, *European Masters in Advanced Robotics + ( EMARO+)*, University of Genoa, Italy / Ecolé Centrale de Nantes, France, A programme of Erasmus Mundus.

2007–2011 **Bachelor of Science - Electrical Engineer - Control Engineering and Robotics**, *Wrocław University of Technology*, Wybrzee Wyspiaskiego 27, PL-50370 Wrocław, *very good / bardzo dobry*.
ISCED 5A, Polish grading system; See supplement to diploma, page 5

2004–2006 **Abitur**, *Berufsbildende Schulen Meppen - gew. und kaufm. Fachrichtungen - Fachrichtung Technik*, Nagelhof 83, D-49716 Meppen,
*3,0 - good / gut*.
ISCED 3A, German grading system

## Engineering project

Title *Metrological properties of a hybrid amplifier*

Supervisor PhD / dr in. Daniel Dusza

Description Design, construction and measurements of a hybrid amplifier with a working frequency within the audible range and a microcontroller based control system.

Grade excellent / celujący

Prizes and Awards Stowarzyszenie Elektryków Polskich - Association of Polish Electrical Engineers
1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011

## Experience

### Vocational

12.07.2010 – 20.08.2010 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein BerSSen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems. Internship for studies at the Wrocław University of Technology.

14.07.2008 – 15.08.2008 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein BerSSen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems.

2 weeks in May 2005 **Computer scientist**, *Ebert Mikrokomputer GmbH*, D-49716 Meppen, Germany.
Computer system installations mostly in the private sector. Customer care on site. Secondary school internship.

---

EMARO+

UNIGE **Projects and assignments:**

- *Localization of a bipod humanoid robot using EKF within ROS framework (**Github repository**)*
- *Study and implementation of developed SLAM algorithms for a biped humanoid robot within ROS framework (**Github repository**)*
- *Mobile robots control: point and posture tracking, Lyapunov control, design and simulation*
- *Mobile robots motion planning using bug, distance based and probabilistic roadmap planning algorithms*
- *Offline and online tuning of EKF in a real world case scenario using landmarks*
- *Technical report on Agriculture in controlled environments*
- *Technical paper review and presentation on FLEXA*
- *Delmia simulations*
- *Design and evaluation of a user interface for a motion analysing application*
- *Basic image processing including edge and object detection, colour based segmentation, optical flow (Lucas-Kanade algorithm) and stereo vision*
- *POSIX compliant publisher / subscriber communication models using pipes and pthreads*
- *Realtime scheduling with RM, PS and EDF*

## Languages

| | | |
|---|---|---|
| Polish | **Native** | |
| English | **B2+** | *Mark: very good / bardzo dobry at SJO WUT* |
| German | **C1** | *100% exam test score, 2nd place in contest at FLD WrUT* |
| Spanish | **A2** | *High school secondary language for 2 years* |
| Italian | **B1** | *Score 25/30, spent 1 year in Genova, Italy within EMARO+ programme* |
| French | **Pending** | *About to spent 1 year in Nantes, France within EMARO+ programme* |

## Computer skills

Operating systems: Extensive experience with Microsoft Windows, GNU/Linux, some experience with hardware specific solutions and FreeBSD

Distributions: Arch Linux (current main OS), OpenSUSE, Funtoo, Gentoo, Linux Mint, Ubuntu, Rasbian

| | | | |
|---|---|---|---|
| Software experience | Various CAD/CAE, office and scientific applications and suites across all media fields including audio, video, graphics, text, design, simulation and engineering | Applications | LibreOffice, Microsoft Office, Altium Designer, Simatic Step 7, CX-Programmer, Delmia, Matlab, Spyder, Pythonxy, Scilab, Proto, Photoshop, Gimp, Inkscape, Ableton Live, Audacity, Geany, TeXstudio, nano |
| Shells / Command line interfaces | Zsh (current main shell), Bash, PowerShell, Command Prompt | Software and community contributions | Proto, Arch Linux Wiki articles (ALSA - Basic Syntax), Arch Linux Forums and packages maintenance, wine, q4wine |

## Programming languages

**General languages and environments**

- *Assembler on various hardware*
- *BASIC on Commodore 64 and various BASIC derivatives on robots*
- *C/C++ in courses, assignments, projects and contributions*
- *Python*
- *ROS - Robot Operating System*

---

**PLC languages**

- *Function block diagram*
- *Ladder logic*

---

**Scripting**

- *Bash / Zsh*
- *Matlab / Scilab / Pythonxy / Spyder*
- *Command Prompt / Batch*
- *Autohotkey*

## Certificates, Licences

| | | |
|---|---|---|
| Electrical qualification | **Operation position** in the area of: **operation**, **maintenance**, **repairs** and **assembly** | *SEP - AoPEE, Wrocław*<br>*Date of exam: 19.05.2010*<br>*Valid for 5 years* |
| Driving licence | **Class A, B** | *Driving licence acquired in Germany in 2005* |

## Organisations, associations and memberships

| | | |
|---|---|---|
| SNS | Academic student association | *Took part in several projects, educational trip and made a publication for the SSC 2009* |
| STRIMER | Wrocław University of Technology | |

## Additional courses

Warsaw

| | |
|---|---|
| 2015 | ABC of enterneurship |
| 2015 | Self presentation and interview skills |

## References

| Title | Name | Position | Workplace |
|---|---|---|---|
| PhD / dr in. | Daniel Dusza ✉ | assistant professor | Wrocław University of Technology, Institute of Electrical Machines, Drives and Measurements |
| Prof PhD / prof. dr hab. in. | Zdzisław Nawrocki ✉ | professor, retired | Wrocław University of Technology, Institute of Electrical Machines, Drives and Measurements |

## Publications

Rzepecki Bartłomiej, Skrzypczyk Ernest, "*Plasma and its applications*", 7th Students' Science Conference, Wrocław, Poland, 2009

## Prizes and Awards

| | |
|---|---|
| Prize | 1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011 |
| Prize | Second place in German language tournament at FLD WrUT |
| Commendation | For best paper presentation at SSC 2009 |
| Aknowledgment | Third place for best group work at SSC 2009 |

## Scholarships

Several scholarships for achievements in studies at the WrUT Faculty of EE

## Interests

**Arts**
- *Graphics, photography, image rendering and manipulation*
- *Music, audio creation and manipulation*
- *Paper model building*

**Philosophy**
- *New socio-economic structures*
- *Consequences of implementation of new technologies*

**Technology**
- *Self-sustainable autonomous vehicles, facilities and infrastructure*
- *Robot interaction*
- *Artificial intelligence*
- *FOSS, especially GNU/Linux*
- *Open Hardware, Open Technology, Open Economy*

## Additional information

| | |
|---|---|
| Mobility | Full mobility within 157 countries, especially across Europe |
| Self-development | Main focus was to shift working environment to and towards FOSS, contribute and exchange with the FOSS community. |